

Worksheet — Exponents and Radicals (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **10**

Instructions: Attempt all 10 questions in your notebook. Show every step of your working for the subjective questions; choose the single best option for MCQs. Answers without working will not receive full marks.

Total time: 50 minutes.

Q1. What is the value of 2^5 ? [1 mark]

- (A) 10
- (B) 16
- (C) 25
- (D) 32

Q2. $a^0 = 1$ for every non-zero real number a . [1 mark]

- (A) True
- (B) False

Q3. $a^{(-n)} = \underline{\hspace{1cm}}$ (in terms of a^n). [1 mark]

Q4. Simplify $a^7 \times a^{(-3)} \div a^2$. [2 marks]

- (A) a^2
- (B) a^4
- (C) a^6
- (D) a^8

Q5. Evaluate $27^{(2/3)}$. [2 marks]

- (A) 6
- (B) 9
- (C) 18
- (D) 27

Q6. Simplify $\sqrt{32} + \sqrt{50} - \sqrt{18}$ to a single surd, showing all steps. [3 marks]

Q7. Rationalise the denominator of $6 / \sqrt{3}$ and express your answer in simplest form. [3 marks]

Q8. Express each of the following with a single fractional exponent: (i) $\sqrt[3]{a^5}$, (ii) $(3\sqrt{a})^4$, (iii) $1 / 5\sqrt{a}$. [4 marks]

Q9. Rationalise the denominator of $1 / (\sqrt{7} - \sqrt{3})$ and verify your answer by checking that the new denominator is rational. [5 marks]

Q10. (a) State the product law of exponents and use it to derive that $a^0 = 1$ for any non-zero real number a . (b) Then derive $a^{-n} = 1 / a^n$ using the result of part (a). [8 marks]

Answer Key

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Q1. What is the value of 2^5 ?

Answer: 32

$$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32.$$

MCQ · EASY · 10-NIOS-MATH-ER-01

Q2. $a^0 = 1$ for every non-zero real number a .

Answer: True

Forced by the product law: $a^0 \times a^n = a^n$ implies $a^0 = 1$.

TRUE_FALSE · EASY · 10-NIOS-MATH-ER-03

Q3. $a^{(-n)} = \underline{\hspace{1cm}}$ (in terms of a^n).

Answer: $1/a^n$

Definition of negative integer exponents.

FILL_BLANK · EASY · 10-NIOS-MATH-ER-03

Q4. Simplify $a^7 \times a^{(-3)} \div a^2$.

Answer: a^2

$$a^7 \times a^{(-3)} = a^4; a^4 / a^2 = a^{(4-2)} = a^2.$$

MCQ · MEDIUM · 10-NIOS-MATH-ER-02

Q5. Evaluate $27^{(2/3)}$.

Answer: 9

$$27^{(1/3)} = 3; 3^2 = 9.$$

MCQ · MEDIUM · 10-NIOS-MATH-ER-04

Q6. Simplify $\sqrt{32} + \sqrt{50} - \sqrt{18}$ to a single surd, showing all steps.

Answer: $\sqrt{32} = 4\sqrt{2}$; $\sqrt{50} = 5\sqrt{2}$; $\sqrt{18} = 3\sqrt{2}$. Total: $4\sqrt{2} + 5\sqrt{2} - 3\sqrt{2} = 6\sqrt{2}$.

Reduce each radical to a multiple of $\sqrt{2}$ by pulling perfect-square factors out.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-ER-06

Q7. Rationalise the denominator of $6 / \sqrt{3}$ and express your answer in simplest form.

Answer: $6/\sqrt{3} \times \sqrt{3}/\sqrt{3} = 6\sqrt{3} / 3 = 2\sqrt{3}$.

Multiply numerator and denominator by $\sqrt{3}$.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-ER-07

Q8. Express each of the following with a single fractional exponent: (i) $\sqrt[5]{a^5}$, (ii) $(3\sqrt[3]{a})^4$, (iii) $1 / 5\sqrt[5]{a}$.

Answer: (i) $\sqrt[5]{a^5} = a^{(5/5)}$. (ii) $(3\sqrt[3]{a})^4 = (a^{(1/3)})^4 = a^{(4/3)}$. (iii) $1/(5\sqrt[5]{a}) = a^{(-1/5)}$.

Use $n\sqrt[n]{a^m} = a^{(m/n)}$ and $a^{(-p)} = 1/a^p$.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-ER-04

Q9. Rationalise the denominator of $1 / (\sqrt{7} - \sqrt{3})$ and verify your answer by checking that the new denominator is rational.

Answer: Multiply numerator and denominator by the conjugate $(\sqrt{7} + \sqrt{3})$: $1/(\sqrt{7} - \sqrt{3}) \times (\sqrt{7} + \sqrt{3})/(\sqrt{7} + \sqrt{3}) = (\sqrt{7} + \sqrt{3}) / ((\sqrt{7} - \sqrt{3})(\sqrt{7} + \sqrt{3})) = (\sqrt{7} + \sqrt{3}) / (7 - 3) = (\sqrt{7} + \sqrt{3}) / 4$. Verify: the new denominator is 4, which is rational. ✓

The conjugate uses the difference-of-squares identity to make the denominator rational.

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Q10. (a) State the product law of exponents and use it to derive that $a^0 = 1$ for any non-zero real number a . (b) Then derive $a^{(-n)} = 1 / a^n$ using the result of part (a).

Answer: (a) Product law: $a^m \times a^n = a^{(m+n)}$. Setting $m = 0$ and any $n \geq 1$: $a^0 \times a^n = a^{(0+n)} = a^n$. Since $a \neq 0$ means $a^n \neq 0$, we may divide both sides by a^n to get $a^0 = 1$. (b) Apply the product law with exponents $-n$ and n : $a^{(-n)} \times a^n = a^{((-n) + n)} = a^0 = 1$ (using part (a)). Dividing both sides by a^n (non-zero) gives $a^{(-n)} = 1 / a^n$.

Both conventions are forced by requiring the product law to remain consistent for all integer exponents.

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